

Recommended Package Installation Guide

Clinical Data Analysis in R — Phase I | Neudata · #ClearDataClearImpact

Run these steps **before Day 1** (or during the Day 1 setup session). You need an internet connection the first time only.

Step 1 — Install R and RStudio

1. **R** (the engine): <https://cran.r-project.org> → download for Windows / macOS → install with default options.
2. **RStudio Desktop** (the workbench): <https://posit.co/download/rstudio-desktop/> → install.
3. Open **RStudio** (not plain R). You should see four panes: Source, Console, Environment, Files/Plots.

> This course was prepared with **R 4.6.0**. Any recent 4.x version works.

Step 2 — Install the course packages

Copy the block below into the RStudio **Console** and press Enter. It installs everything used across the five days in one go.

```
install.packages(c(
  # --- core: import, wrangle, visualise ---
  "tidyverse",      # dplyr, ggplot2, readr, stringr, tibble, lubridate ...
  "readxl",         # read Excel (.xlsx) files
  "lubridate",      # working with dates
  "janitor",        # quick cleaning helpers (clean_names, tabyl)

  # --- tables & reporting ---
  "gtsummary",      # publication-ready Table 1 and regression tables
  "gt",             # rendering / exporting tables
  "broom",          # tidy model output (odds ratios, CIs)
  "broom.helpers",  # required by gtsummary for tbl_regression()

  # --- statistics & diagnostics ---
  "car",            # VIF and regression diagnostics
  "pROC",           # ROC curve / AUC
  "scales",         # nice axis formatting
))
```

Installation can take several minutes. Windows and macOS receive pre-compiled binaries, so no extra build tools are needed.

Step 3 — Confirm everything loaded

Run this check. Every line should print TRUE.

```
pkgs <- c("tidyverse","readxl","lubridate","janitor","gtsummary",
          "gt","broom","broom.helpers","car","pROC","scales")
for (p in pkgs) cat(sprintf("%-15s %s\n", p, requireNamespace(p, quietly = TRUE)))
```

Step 4 — Set up the project

1. Download/unzip the **Course** folder supplied by the instructor.
 2. In RStudio: **File** → **New Project** → **Existing Directory** → choose the Course folder.
 3. Working from this project means all paths like `Data/hypertension_phc_raw.csv` "just work" on any computer.
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Optional extras

Package	Why you might want it
rmarkdown / quarto	one-click reproducible reports (PDF/Word/HTML)
webshot2 + chromote	export gt tables to PNG/PDF (needs Chrome installed)
here	robust relative paths in larger projects
flextable / officer	send tables straight to Word/PowerPoint

Troubleshooting

- **"there is no package called X"** → you skipped Step 2, or `library(X)` is spelled differently from `install.packages("X")`.
- **Table export to PNG fails / "Chrome not found"** → install Google Chrome, or export tables as `.html/.csv` instead (the course scripts fall back automatically).
- **Corporate network blocks CRAN** → set a mirror: `options(repos = c(CRAN = "https://cloud.r-project.org"))` then retry, or ask IT for an approved mirror.
- **Slow install** → install in smaller groups (core first, then reporting, then stats).